FarmPulse AI

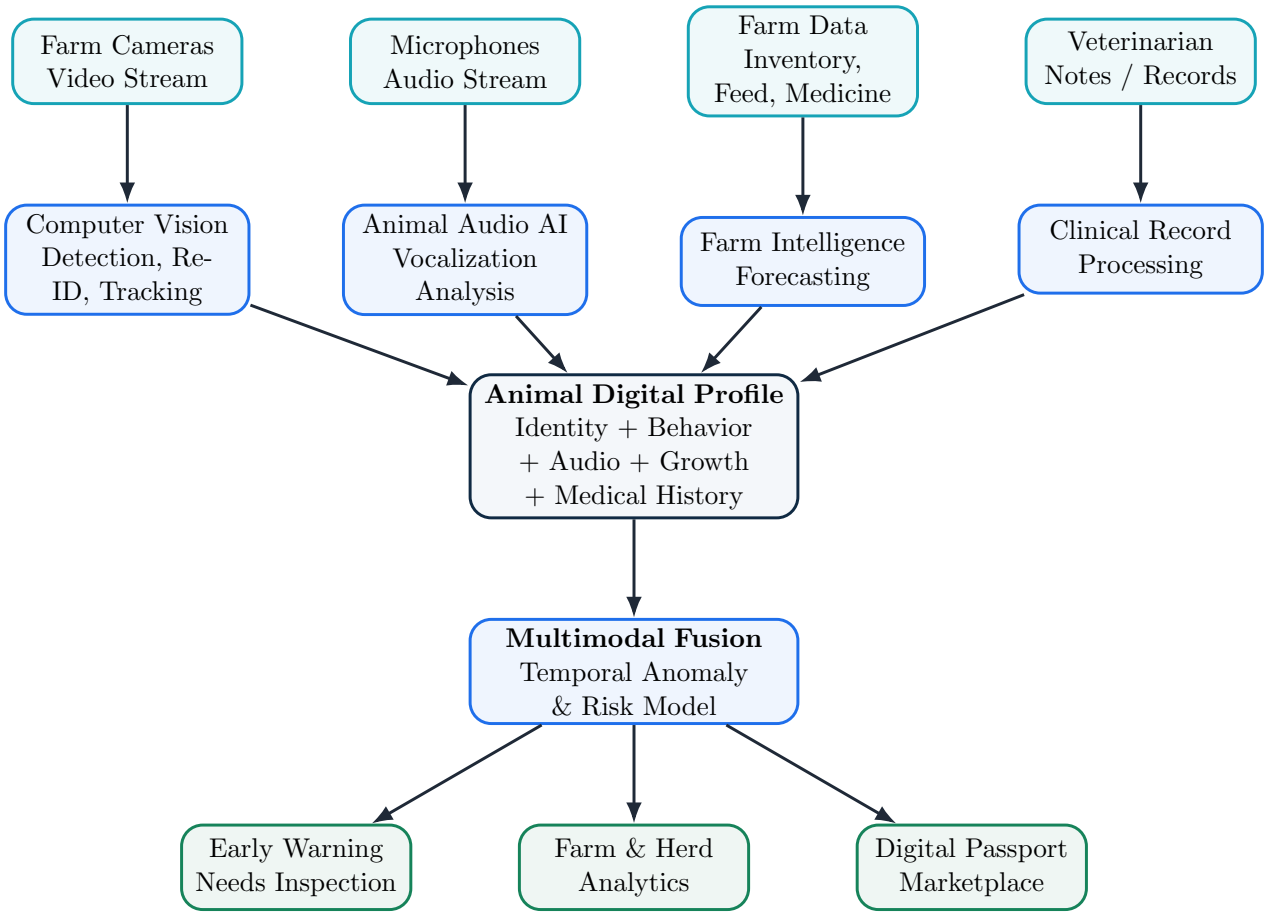
Multimodal AI for Animal Monitoring, Early Risk Detection,
Farm Management, and Digital Animal Profiles

1. Project Idea

The proposed system is an AI-powered livestock platform for cattle and buffalo farms. Its core objective is to continuously understand each animal using cameras, microphones, historical records, and farm data. Instead of only detecting an animal in a frame, the platform builds a long-term digital profile for every animal and identifies meaningful changes in behavior, activity, feeding, drinking, growth, and vocalization.

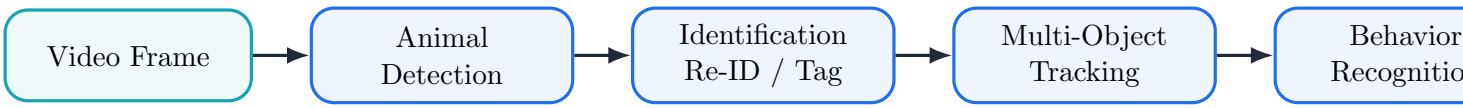
The system is designed as a **decision-support and early-warning platform**. It should flag abnormal patterns and recommend veterinary inspection rather than claim a medical diagnosis from video or audio alone.

2. High-Level System Flow



3. Core Computer Vision Pipeline

The camera subsystem should go beyond simple YOLO detection. The research value is in maintaining the identity of each animal and understanding its behavior over time.



Possible recognized behaviors include:

- walking, standing, lying, eating, drinking, and resting;
- isolation from the herd or unusual inactivity;
- repeated movement or other abnormal behavior, if supported by reliable labels;
- feeding and drinking visit frequency and duration.

For every animal, the platform can create a behavioral baseline such as daily movement, feeding duration, drinking frequency, and lying time. Changes from that baseline become inputs to the anomaly model.

4. Animal Vocalization and Distress Audio Model

A microphone network can provide a second modality. The objective is not to diagnose a disease from sound alone, but to detect **abnormal vocalizations or acoustic patterns associated with distress or health-related changes**, depending on the available ground-truth dataset.



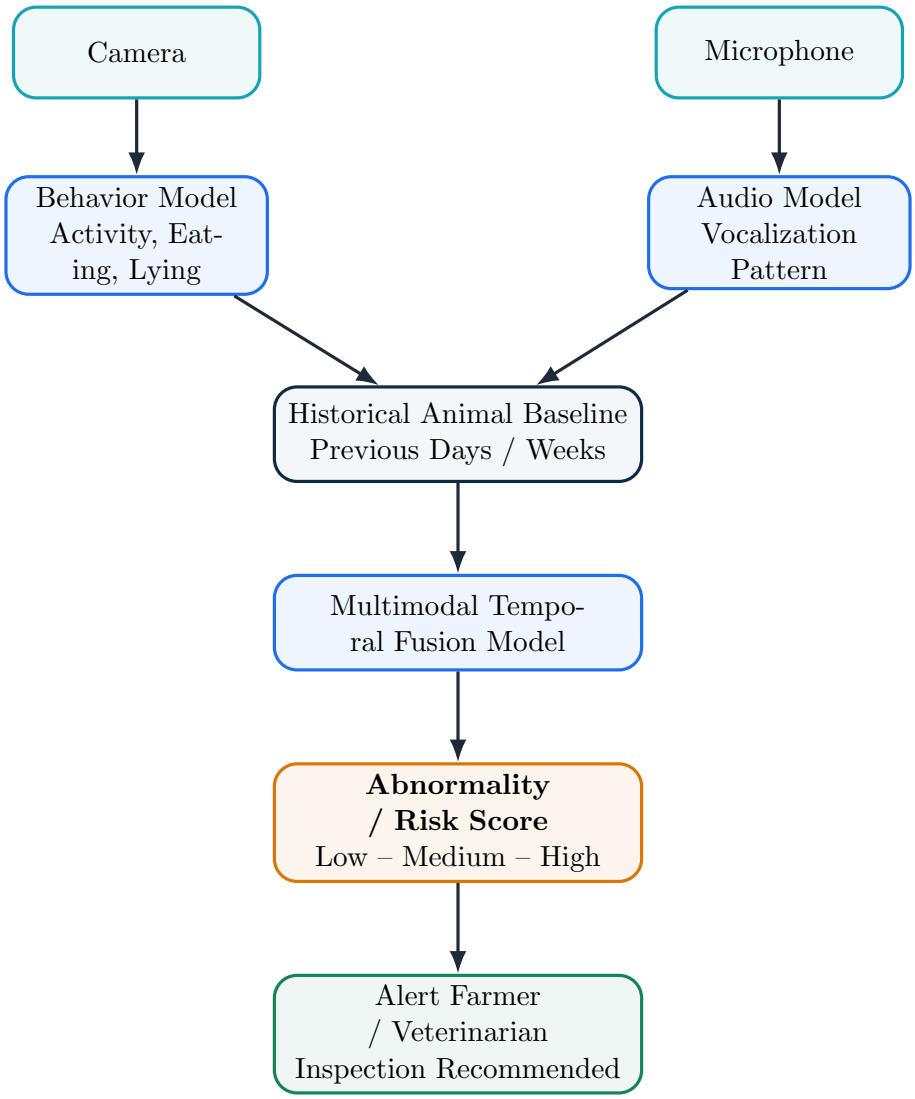
A from-scratch experimental architecture can be trained from random initialization:

$$\text{Audio} \rightarrow \text{Spectrogram} \rightarrow \text{CNN} \rightarrow \text{BiGRU} \rightarrow \text{Attention} \rightarrow \text{Classification}$$

Potential targets, subject to dataset quality, include normal vocalization, distress, coughing/respiratory sounds, hunger-related calls, or other labeled acoustic events.

5. Multimodal Early-Warning Flow

The strongest version of the project combines visual behavior and animal audio instead of making decisions from a single signal.



Example: if an animal's movement and feeding time fall significantly below its personal baseline while its vocalization pattern becomes abnormal, the combined model can assign a higher risk score and request inspection.

6. Animal Digital Profile

Every animal should have a persistent ID. Identification may use visual re-identification, an ear tag/RFID integration, or a hybrid approach.

Animal C104

Breed / Age / Sex
Weight and Growth History
Vaccination History
Medication and Vet Records
Daily Activity Timeline
Feeding and Drinking Trends
Audio Abnormality History
Risk Alerts
Marketplace Status

This profile becomes the connection between AI monitoring, farm operations, veterinary records, analytics, and the optional marketplace.

7. Farm Operations

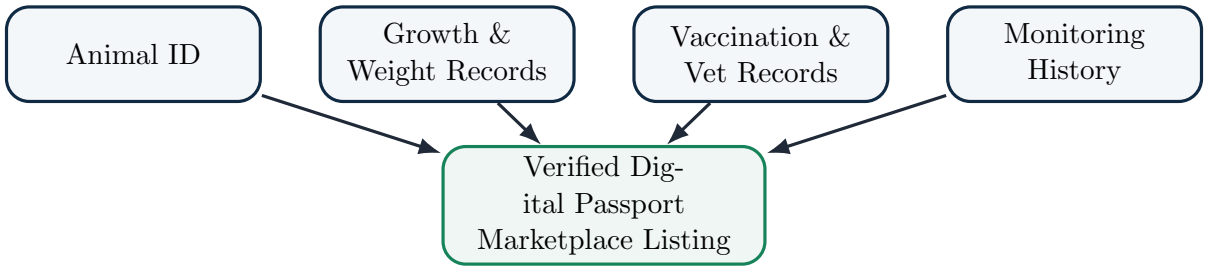
Not every feature needs to be an AI model. Standard software modules can handle:

- medication and vaccination schedules;
- veterinary check-in/check-out and visit records;
- feed, medicine, vaccine, and equipment inventory;
- notifications and missed-treatment alerts;
- animal registration and farm user management.

AI can be added where it provides measurable value, for example feed-demand forecasting, growth forecasting, or anomaly detection.

8. Smart Marketplace and Digital Animal Passport

The marketplace should remain connected to the core system. Instead of a generic listing, an animal can have a traceable digital passport containing verified farm records.



9. Recommended AI Work for Four Students

Student	AI Problem	Main Output
AI 1	Animal detection, individual identification / Re-ID, and tracking	Animal ID and trajectory over time
AI 2	Video behavior/action recognition	Eating, drinking, walking, standing, lying, and other supported behaviors
AI 3	Animal vocalization and abnormal/distress audio analysis	Audio event class and abnormality probability
AI 4	Multimodal temporal fusion and early-risk prediction	Animal anomaly/risk score using vision, audio, and historical behavior

This division keeps all four students focused on the same central research question while giving each student a separate dataset, model, experiments, metrics, and final output.

10. Recommended Scope

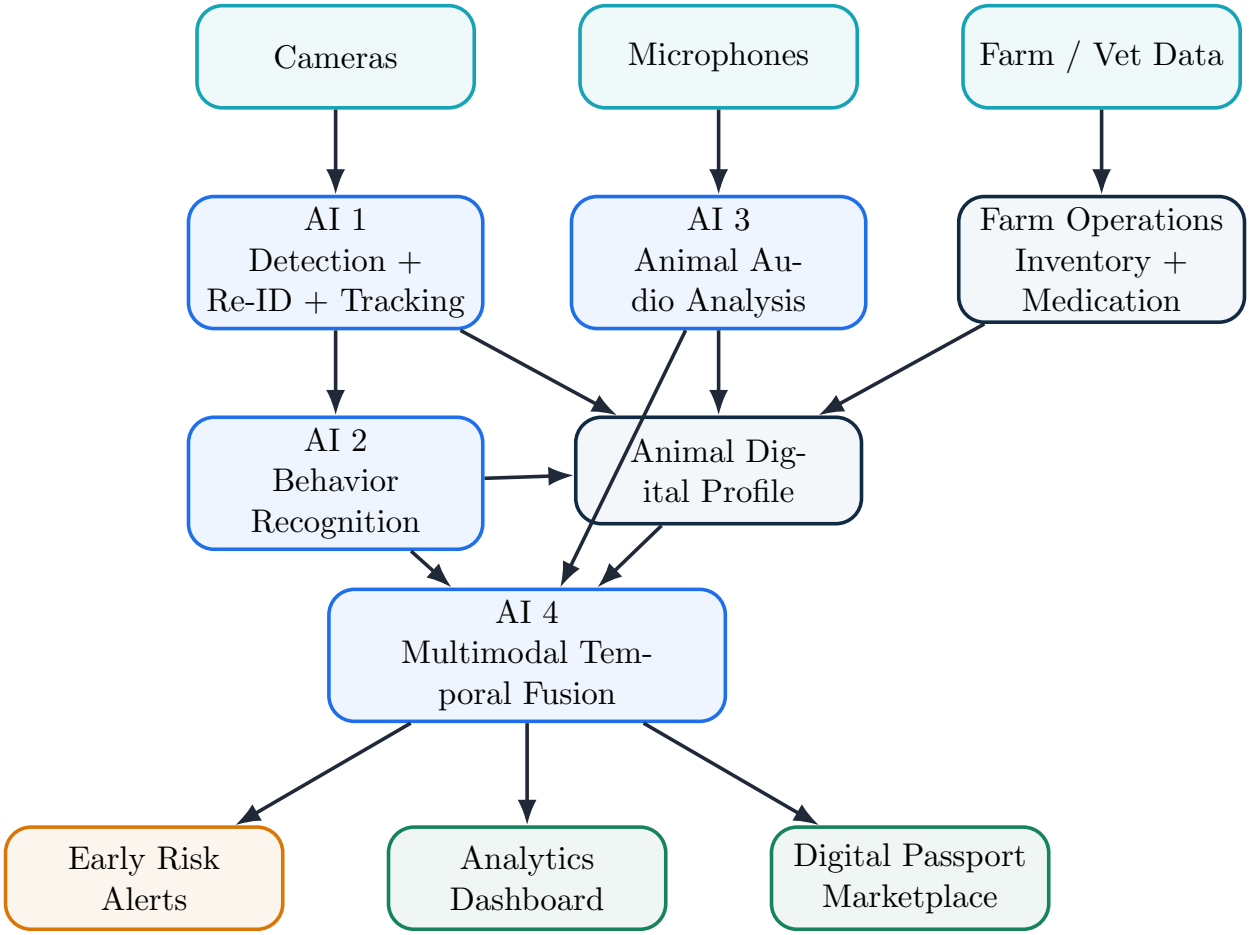
For the first version, the project should focus on **one livestock family, preferably cattle/buffalo**, rather than attempting cattle, buffalo, poultry, and other animals simultaneously. Behavior classes, body structure, audio characteristics, and available datasets differ substantially across species.

A practical first scope is:

Cattle Monitoring → Individual Tracking → Behavior Recognition → Audio Analysis → Multimodal Early Warning

After validating this pipeline, inventory forecasting, weight estimation, veterinary speech notes, digital passports, and marketplace functionality can be added as extensions.

11. Final Architecture



12. Core Research Question

Can continuous visual behavior and animal vocalization analysis, combined with each animal's historical baseline, provide useful early warnings of abnormal livestock conditions and help farmers identify animals that require inspection?

13. Dataset Allocation

The following datasets are the best-fit accessible datasets for the four core AI research tasks in FarmPulse AI.

Student	Selected Dataset(s)	Research Task
AI 1	MultiCamCows2024 + OpenCows2020	Animal detection, multi-object tracking, individual identification, and cattle Re-ID across cameras.
AI 2	CBVD-5 + XGain Cattle Behavior Dataset	Temporal behavior recognition including eating/feeding, drinking, standing, lying, grazing, rumination, and movement-related activities.
AI 3	CowRumIA + Daylong Cow Acoustic Dataset	Cattle bioacoustic intelligence: stress/calm vocalization analysis, grazing/rumination acoustic events, and abnormal sound-pattern detection.
AI 4	Outputs/features from AI 1–3 + historical animal records	Multimodal temporal fusion combining identity, behavior, audio, and historical baselines to produce an abnormality/risk score and early-warning signal.

Dataset access:

- MultiCamCows2024: https://data.bris.ac.uk/data/dataset/2inu67jru7a6821kkgehxg3cv2
- OpenCows2020: https://data.bris.ac.uk/data/dataset/10m32xl88x2b61zlkkgz3fml17
- CBVD-5: https://www.kaggle.com/datasets/fandaoerji/cbvd-5cow-behavior-video-dataset
- XGain Cattle Behavior Dataset: https://zenodo.org/records/15688349
- CowRumIA: https://zenodo.org/records/20406266
- Daylong Cow Acoustic Dataset: https://www.nature.com/articles/s41597-023-02673-3